Quantum mechanics

A. Problems on de-broglie's hypotheses

Formulae used

1.] $\lambda = \frac{h}{mv}$

Where,

λ =wavelength of light

m =mass

v = velocity

h = plank constant

2.] $\lambda = \frac{h}{p}$

Where,

p = momentum

and p =mv

3.] $\lambda = \frac{h}{\sqrt{2mE}}$

Solved example

Q1. A bullet of mass 40gms and an electron both travel at the velocity of 1100m/sec. What wavelengths can associated with them? Why the wave nature of bullet is not revealed through diffraction effect?

Soln:

Mass of an electron = 9.1×10^{-31} kg

Mass of bullet = 40×10^{-3}Velocity of bullet=1100m/sec

Let wavelength associated with bullet be λ1

λ1 = h/mv

$$=\frac{6.63\times10^{-34}}{40\times10^{-3}\times1100}$$

$$= 1.5\times10^{-35}\text{m}$$

Let wavelength associated with electron be λ2

λ2= h/mv

$$=\left(\frac{6.63\times10^{-34}}{9.1\times10^{-31}\times1100}\right)$$

$$= 6.62\times10^{-7}\text{m}$$

λ2= *6620A*

Q2.] Electrons accelerated through 100V are reflected from crystal. What is the glancing angle at which the first reflection occurs? Lattice spacing=2.15A

Soln.:

V= 100V

Wavelength associated with electrons is

$$\lambda= \frac{12.26}{\sqrt{V}}$$

$$\lambda=\frac{12.26}{\sqrt{100}}$$

=1.226A

Now using Bragg's law for first order:

$\lambda = 2d\sin\theta$

$1.226A = 2x2.15xA^0 x\sin\theta$

$\theta = 16.56.$

Q3.] Calculate the kinetic energy of an electron whose de broglie wavelength is 5000A.

Soln:

Mass of electron = 9.1×10^{-31} kg

$h = 6.625x10^{-34}$J.sec

K. E =?

Using formula,

$\lambda = h/\sqrt{2mE}$

$E = \frac{h^2}{2m\lambda^2}$

$= \left(\frac{(6.625\,X\,10^{-34})^2}{2\,X\,9.1\,X\,10^{-31}\,X\,(5000\,X\,10^{-10})^2}\right)$

$= 9.6378x10^{-25}$ J = $6.023x10^{-6}$eV

Q4.] Calculate the velocity and the de broglie wavelength of α particles of energy 1keV.

Given: Mass of α particle= $6.68x10^{-27}$

Soln:

E = 1keV=$1x10^3x1.6x10^{-19}$ J

$\lambda = h/\sqrt{2mE}$

$$=\frac{6.63\,X\,10^{-34}}{\sqrt{2\,X\,6.68\,X\,10^{-27}\,X\,1\,X\,10^{3}\,X\,1.6\,X\,10^{-19}}}$$

$= 4.534x10^{-13}m$

Now,

λ=h/mv

$$V=\frac{6.63\,X\,10^{-34}}{6.68\,x\,10\wedge-27\,x\,4.534\,x\,10\wedge-13}$$

$V=218.9x10^{3}m/sec$

Q5.] What is the deBroglie wavelength of an electron with 13.6 eV of kinetic energy? What is the deBroglie wavelength of an electron with 10 MeV of kinetic energy?

Answer

13.6 eV is much less than $mc^2 = 0.511 MeV$ so this is non-relativistic.

$$\begin{aligned}
\frac{p^2}{2m} &= 13.6 \\
\frac{p^2c^2}{2mc^2} &= 13.6 \\
pc &= \sqrt{2mc^2(13.6)} \\
\lambda &= \frac{h}{p} = \frac{2\pi\hbar}{p} = \frac{2\pi\hbar c}{pc} = \frac{2\pi\hbar c}{\sqrt{2mc^2(13.6)}} \\
&= \frac{2\pi(1973\ eV\AA)}{\sqrt{2(0.511\times10^6\ eV)(13.6\ eV)}} \approx 3.33\ \AA
\end{aligned}$$

10 MeV is much bigger than for an electron so it is super-relativistic and we can use .

Q6.] What is the wavelength of an electron moving at 5.31 x 10 ^{6} m/sec?

Given: mass of electron = 9.11 x 10-31 kg

h = 6.626 x 10-34 J•s

Soln:

de Broglie's equation is

λ = h/mv

$$= \frac{6.626\,x\,(10)\wedge(-34)}{9.11\,x\,(10)\wedge(-31)\ \ x\,5.31\,x\,(10)\wedge 6}$$

λ = 1.37 x 10-10 m

λ = 1.37 Å

Q7.] What us the de Broglie wavelength of 75g ball moving with speed of 42m/s?

Soln:

λ = h/p

= h/mv

λ =6.63x(10)^(-34)/((64x10)^(-3) x42)

λ = 2.1x10-34m

Unsolved example

Q1.] Determine the wavelength of an electron moving with the speed of $5.41x10^{5}$ m/s.
(Ans1.37x10 $^{-9}$ m)

Q2.] Determine the wavelength of the particle. Consider an electron with velocity at 10^7 cm/ sec (Ans $7.27x10^{-11}$ m)

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Q3.] What is de broglie wavelength if a photon with momentum of 4.4x10-34kg.m/s?
(Ans1.5m)

Q4.] The momentum if photon A is pa and momentum of photon B is pb. If pb= 1/3pa then what us the relation the wavelengths?
(Ans=λa=(1/3)λb)

Q5.] Calculate the de Broglie wavelength of a golf ball whose mass is 40 grams and whose velocity is 6 m/s.
(Ans=$2.76x10^{-33}$m)

Q6.] What is the frequency of the photon whose energy is 66.3eV?
(Ans=$1.6x10^{16}$hz)

Q7.] A photon of energy $6x10^{-20}$ J has linear momentum of $2x10^{-30}$kg.m/s. Verify this statement.

B. Problems on uncertainty principle

Formulae used

I. $\Delta x \Delta p_x \geq \hbar$

Where $\Delta x \rightarrow$ uncertainty in position of particle

$\Delta p_x \rightarrow$ uncertainty in momentum of particle

a. Also $\Delta p_x = m\,\Delta v_x$

$\Delta v_x \rightarrow$ uncertainty in velocity of particle

II. $\Delta E \Delta t \geq \hbar$

Where $\Delta E \rightarrow$ uncertainty in energy

$\Delta t \rightarrow$ uncertainty in time spent in excited st

Solved examples

1. An electron has a speed of 400 m/s with uncertainty of 0.01%. Find the accuracy in its position.

   Soln:

   Momentum of electron = p = mv

   $= 9.11 \times 10^{-31} \times 400 = 3.644 \times 10^{-28}$ kg.m/sec

   $\therefore$ using Heinsenberg's uncertainty formula

   $$\Delta x \Delta p_x \geq \hbar$$

   $$\Delta x \geq \frac{h}{2\pi \Delta p_x}$$

   $$\geq \frac{6.63 \, x \, 10^{-34}}{2\pi \ x \ 3.644 \ x \ 10^{-32}}$$

   $$\geq 2.895 \times 10^{-3} \text{ m}$$

2. An electron is confined in a box of length 10^{-8} m. calculate minimum uncertainty in its velocity.

Soln:

As $\Delta x \Delta p_x \geq \hbar$ $\therefore \Delta x_{max} \Delta p_{xmin} \geq \hbar$

Now box has length 10^{-8}m

$$\therefore \Delta x_{max} = 10^{-8}$$

$$\therefore \Delta p_{xmin} = \frac{h}{2\pi \ x 10^{-8}} = \frac{6.63 \ x \ 10^{-34}}{2\pi \ x \ 10^{-8}}$$

$$\therefore \Delta p_{xmin} = m \, \Delta v_{xmin}$$

$$\therefore \Delta v_{xmin} = \Delta p_{xmin}/m$$

$$= \frac{1.5066 \ x \ 10^{-26}}{2\pi \ x \ 10^{-31}} = 11.595 \text{ x } 10^{3} \text{ m/sec}$$

3. The speed of electron is measured to within an uncertainty of 2 x 10^{-8}m/s. what is the minimum space required by electron to be confined in an atom?

Soln:

Using eq of uncertainty

$$\Delta x . \Delta p_x \geq \frac{h}{2\pi}$$

$$\therefore \Delta x \geq \frac{h}{2\pi \Delta p_x}$$

$$\geq \frac{h}{2\pi \Delta v_x} \geq \frac{6.63 \ x \ 10^{-34}}{2\pi \ x \ 9.1 \ x \ 10^{-31} x \ 2 \ x 10^{4}}$$

$$\geq 0.579 \text{ x } 10^{4} \text{ m}$$

4. The inherent uncertainty in the measurement of time spent by a nucleus in the excited is found to be 1.4 x 10^{-10} sec. Estimate that results in its energy in excited state.

Soln:

Using equation $\Delta E . \Delta t \geq \hbar$

$$\Delta E \geq \frac{h}{2\pi\Delta t}$$

$$\geq \frac{6.63 \ x \ 10^{-34}}{2\pi \ x \ 1.4 \ x \ 10^{-10}}$$

$$\geq 7.537 \text{ x } 10^{-25} \text{ J}$$

$$\geq \frac{7.537 \ x \ 10-25}{1.6 x \ 10^{-19}} \text{ eV}$$

$$\geq 4.71 \text{x} 10^{-8} \text{ eV}$$

5. An electron moves in the x-direction with a speed of $1.88 \text{x} 10^{-8}$ m/s.
   If this speed measured to a precision of 1% with what precision can you simultaneously measure its position?
   Soln:

   Uncertainty in measurement of speed

   $$\Delta v = \frac{1.88 \ x \ 10^{6}}{100} = 1.88 \text{ x } 10^{4} \text{ m/s}$$

   Accoding to Heisenberg's uncertainty principle

   $$\Delta x . \Delta p \geq \hbar$$

   $$\geq \frac{h}{2\pi}$$

   $$\Delta x = \frac{h}{2\pi \Delta p}$$

   $$\Delta x = \frac{h}{2\pi m \Delta v}$$

   $$\Delta x = \frac{6.63 \ x \ 10^{-34}}{2\pi \ x \ 9.1 \ x 10^{-31} \ x \ 1.88 \ x \ 10^{4}}$$

   $$\Delta x = 6.16 \text{ x } 10^{-9} \text{ m} = 6.16 \text{ nm}$$

6. A hydrogen atom has diameter of 0.53 A^0. Estimate the minimum energy an electron can have in this atom.
   Soln:

   (a) Calculation of minimum momentum of an electron

   According to heisenberg's uncertainty principle

   $$\Delta x . \Delta p \geq \hbar$$

$$\Delta x_{max}.\Delta p_{min} = \frac{h}{2\pi}$$

$$\Delta p_{min} = \frac{h}{2\pi \Delta x_{max}}$$

$$= \frac{6.63\, x\ 10^{-34}}{2\pi\ x\ 0.53\ x\ 10^{-10}}$$

$$\Delta p_{min} = 9.954 \text{ x } 10^{-25} \text{ kgm/s}$$

(b) Calculation of minimum of energy of electron

$$\text{Kinetic energy (E)} = \frac{1}{2} m v^2 = \frac{p^2}{2m}$$

$$\text{Kinetic energy } (E_{min}) = \frac{{p_{min}}^2}{2m}$$

$$= \frac{(9.94\, x\, 10^{-25})^2}{2\ x\ 9.1\ x\ 10^{-31}}$$

$$= 5.444 \text{ x } 10^{-19} \text{ J}$$

Unsolved problems

1) If the uncertainty in position of an electron is $4x10^{-10}$ m. Calculate uncertainty in its momentum.
(ans $2.62x10^{-25}$ kg m/sec)
2) A bullet of mass 0.050 kg is moving with a velocity of 800m/s. The speed of bullet measured accuracy 0.01%. Calculate the accuracy with which the position of the bullet can be located.
(ans: $2.62x10^{-32m}$)

3) Calculate the minimum uncertainty in measurement of momentum of an electron if the uncertainty in locating it is 1 A°.
(ans: 1.055×10^{22} kg m/sec)

4) The uncertainty in the location of a particle is equal to its de Broglie wavelength. Calculate the uncertainty in its velocity.

5) An electron and a 150 gm baseball are travelling at 220 m/sec measured to an occupy of 0.065 %. Calculate the uncertainties in the position of each of the bodies. Compare and comment
on the results.
(ans: uncertainty of electron is greater than baseball)

6) An electron has a momentum 5.4×10^{-26} kg-m/sec with an accuracy of 0.05%. Find the minimum uncertainty in the location of electron. (ans: 3.9 μm)

C. Problems On Group And Phase Velocity

Formulae used

I. $v_{group} = v$

II. $(v_{group})(v_{phase}) = c^2$

Where,

$v \rightarrow$ velocity of the particle

$v_g / v_{group} \rightarrow$ velocity of the wave packet or group velocity

$v_p / v_{phase} \rightarrow$ velocity of the wave or phase velocity

SOLVED examples

1. A fast moving neutron has de-broglie wavelength 2×10^{-12} m associated with it. Find the following:
   (a) Kinetic energy
   (b) Phase and group velocity
   ($m_{neutron} = m_n = 1.675 \times 10^{-27}$ kg)

   Soln:

   $$p = \frac{h}{\lambda} = \frac{6.63 \ x \ 10^{-34}}{2 \ x \ 10^{-12}} = 3.32 \text{ x } 10^{-22}$$

   $$\therefore \ KE = p^2/2m = \frac{(3.32 \, x \, 10^{-22})^2}{2 \ x \ 1.675 \ x \ 10^{-27}} = 3.280 \text{ x } 10^{-17}$$

   since the velocity of the particle is same as the group velocity, the group velocity can be given as,

   $$v_{group} = p/m = \frac{3.32 \ x \ 10^{-22}}{1.675 \ x \ 10^{-27}} = 1.98 \text{ x } 10^5$$

   now using the relation

   $$(v_{group})(v_{phase}) = c^2$$

   $$\therefore \ v_{phase} = \frac{c^2}{v_{group}} = \frac{(3 \, x \, 10^8)^2}{1.98 \, x \ 10^5} = 4.55 \text{ x } 10^{11} \text{ m/sec}$$

2. A particle of mass 1.57×10^{-30} kg and kinetic energy 80eV. Find the de Broglie wavelength ,group and phase velocity of that wave.
   Soln:

   $$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.63 \, x \, 10^{-34}}{\sqrt{2 \, x \, 1.157 \, x \, 10^{-30} \, x \, 80 \, x \, 1.6 \, x \, 10^{-19}}} = 1.128 \times 10^{-10} \text{m}$$

   Now,

   $$V_{group} = \frac{p}{m} = \frac{h}{\lambda . m} = \frac{6.63 \ x \ 10^{-34}}{1.218 \ x \ 10^{-10} \ x \ 1.157 \ x \ 10^{-30}}$$

   $$V_{group} = 4.7 \times 10^6 \text{ m/sec}$$

$V_{phase} = \frac{c^2}{v_{group}} = \frac{(3 x 10^8)^2}{4.7 x 10^6} = 1.1913x10^{10}$ m/sec

Unsolved problems

1. Find phase and group velocities of electron whose de Broglie wavelength is 1.2 A^0.
   (v_p=3.036x10^6 m/s, v_p=6.06x10^6 m/s)
2. The electrons which are at rest accelerated through a potential difference of 250 V. calculate
   I. The velocity of electron
   II. Phase velocity of electron
   III. Debroglie wavelength , momentum, wave number
   (ans: v=9.376x10^6m/s , v_p=9.59x10^9 m/s , λ=7.78x10^{-11}m, p= 8.532x10^{-24} kg m/s , n=1.28x10^{10}/m)

D. Problems on Schrodinger's equation

Formula used

I. $E_n = \frac{P_n^{\ 2}}{2m}$, $E_n = \frac{n^2 h^2}{8 mL^2}$

II. $P(x)=\int_0^L [\varphi(x)]^2 dx$

III. $\Psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$

Where,

E_n= energy states of particle

n = no of state

L= width of the area in which particle in enclosed

P(x)=probability of finding particle within the area

$\Psi_n(x)$=wave function of particle

Solved examples

1. A particle is moving in an one dimensional box is described by a wave function.

$\Psi(x) = \sqrt{3}x$ for 0<x<1

= 0 elsewhere

Find the probability of finding the particle within the interval (0,1/2)

Solution:

We know,

$P(x)=\int_0^L [\varphi(x)]^2 dx$

$P(x)=\int_0^{1/2} 3x^2 dx = 1/8$

P(x)=1/8

2. An quantum particle confined to an one dimensional box of width ‘a’ is in its first excited state. What is the probability of finding the particle over an interval of ‘a/2’ at the center of the box?

Solution:

We know,

P(x)=$\int_0^L [\varphi(x)]^2 dx$

Ψ_n(x) =$\sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L}$

$\therefore$here we have,

P_n(x) = $\frac{2}{a}.\int_{a/4}^{3a/4}(\sin\frac{2\pi x}{a})^2$dx

= $\frac{2}{a}.\int_{a/4}^{3a/4}\frac{1}{2}(1-\cos\frac{4\pi x}{a})$dx

= 0.5

Hence, probability=0.5

3. Find the values of momentum and energy of a neutron in one dimensional box with impenetrable wall. Given,m_n =1.67x10^{-27} kg,L=1A

Solution:

Formula: $E_n=\frac{P_n^2}{2m}$,$E_n=\frac{n^2h^2}{8mL^2}$

n=1,2,3……..

Calculations:

$$E_n=\frac{n^2h^2}{8mL^2}$$

$$= \frac{n^2(6.63\,X\,10^{-34})^2}{8\,X\,6.62\,X\,10}$$

$$= \frac{n^2(6.63\,X\,10^{-34})^2}{(8\,X\,6.62\,X\,10^{-27}\,X\,10^{-10})^2}$$

$$E_n = n^2\,X\left(3.32\,X\,10^{-21}\right)J$$

$E_1 = (3.32\,X\,10^{-21})$ J

$E_2 = (13.28\,X\,10^{-21})$ J

$E_3 = (29.88\,X\,10^{-21})$ J and so on

$$\frac{p^2{}_n}{2m} = \frac{n^2h^2}{8mL^2}$$

$$p_n = \frac{nh}{2L} = \frac{nX\,6.63\,X\,10^{-34}}{2\,X\,10^{-10}}$$

$=n\ (3.315\text{x}10^{-24})$ kg m/sec

$p_1 = 3.315\text{x}10^{-24}$ kg m/sec

$p_2 = 6.63\text{x}10^{-24}$ kg m/sec

$p_3 = 9.5\text{x}10^{-24}$ kg m/sec and so on

Unsolved

1. Calculate the energy of the lowest three levels for an electron in a square well of width $2A^0$.
   (ans: $E_1=15.1\text{x}10^{-19}$J
   $E_2=60.4\text{x}10^{-19}$J

$E_3=135.9x10^{-19}J$)

2. Find the energy of an electron moving in one dimension in an infinitely high potential bol of width $1A^0$. Also calculate the energy difference between the ground state and the first excited state.
   (ans: energy=$n^2x6.038x10^{-18}J$
   Energy difference=$18.114x10^{-18}J$)
3. Find the lowest energy of an neutron within a nucleus of dimension 10^{-14} m. (the mass of neutron= $1.67x10^{-27}kg$)
   (Ans:$3.29x10^{-13}J$)
4. The lowest energy of an electron trapped in a potential well in 38eV.
   Calculate the width of well.
   (Width=0.9965 $A^{0)}$
5. A particle is confined to an one dimensional box of width 25 A^0 and infinite height. calculate the probability of finding the particle within an interval of $5A^0$ at this center of the box in its least energy state.
   (probability=20%)